

AS Chemistry — Topic 1 Atomic Structure (Exam · Paper B)

Teacher mark scheme · CIE 9701 · 33 marks

This file is for teachers only. CIE IDs and letters are listed so the paper can be remapped. Student **main.pdf** has questions only.

Section A — MCQ key

Q	Letter	CIE ID	Paper	vs A
A1	D	cie-9701-2024-mj-p11-q4	2024 MJ P11 Q4	new
A2	C	cie-9701-2021-on-p12-q3	2021 ON P12 Q3	new
A3	D	cie-9701-2025-mj-p13-q23	2025 MJ P13 Q23	new
A4	C	cie-9701-2025-mj-p12-q14	2025 MJ P12 Q14	shared
A5	D	cie-9701-2025-on-p11-q40	2025 ON P11 Q40	shared
A6	B	cie-9701-2025-on-p11-q1	2025 ON P11 Q1	new
A7	D	cie-9701-2025-mj-p11-q14	2025 MJ P11 Q14	new
A8	C	cie-9701-2021-mj-p11-q2	2021 MJ P11 Q2	new
A9	C	cie-9701-2021-mj-p11-q7	2021 MJ P11 Q7	shared
A10	C	cie-9701-2023-on-p12-q2	2023 ON P12 Q2	new
A11	C	cie-9701-2025-mj-p11-q4	2025 MJ P11 Q4	shared
A12	A	cie-9701-2021-on-p11-q3	2021 ON P11 Q3	new
A13	A	cie-9701-2022-fm-p12-q1	2022 FM P12 Q1	new

Letters only: A1 D · A2 C · A3 D · A4 C · A5 D · A6 B · A7 D · A8 C · A9 C · A10 C · A11 C · A12 A · A13 A

B1 · 2025 MJ Paper 22 Question 2 (10 marks)

- (i) [2] M1 average (weighted) mass of an atom / the isotopes. M2 compared to the unified atomic mass unit / $^{12}\text{C} = 12$ exactly cie-9701-2025-mj-p22-q2a-i
- (ii) [2] M1 abundance of third isotope = 2.1%. M2 A_r rearrangement $\Rightarrow$ relative isotopic mass **56.9** cie-9701-2025-mj-p22-q2a-ii
- 1 **1** pair (in 3d of Fe^{2+}) cie-9701-2025-mj-p22-q2b
- 1 spherical s orbital (circle) cie-9701-2025-mj-p22-q2c
- 2 electrons **23**; neutrons **30** cie-9701-2025-mj-p22-q2d
- 2 M1 smaller / decreases. M2 loss of outer shell **or** greater nuclear attraction after electron loss cie-9701-2025-mj-p22-q2e

B2 · 2023 MJ Paper 21 Question 1(b)–(d) (6 marks)

Deleted 1(a) Table 1.1. Paper letters: old (b)(c)(d) $\rightarrow$ new (a)(b)(c).

- 1 **1s** cie-9701-2023-mj-p21-q1b
- 1 $\text{Te}(\text{g}) \longrightarrow \text{Te}^+(\text{g}) + \text{e}^-$ cie-9701-2023-mj-p21-q1c
- (i) [2] Any two: increased nuclear attraction; decreased shielding (IE 4 to 5); spin-pair repulsion (IE 1 to 2). Not “increase in nuclear charge” cie-9701-2023-mj-p21-q1d-i
- (ii) [2] M1 general increase. M2 jumps 4 to 5 and (largest) 6 to 7 cie-9701-2023-mj-p21-q1d-ii

B3 · 2025 MJ Paper 24 Question 6(c) (4 marks)

- 2 M1 $n = 100 \times 11.0 / (1.1 \times 100)$. M2 **10** cie-9701-2025-mj-p24-q6c-i
- 1 $\text{C}_{10}\text{H}_{18}$ cie-9701-2025-mj-p24-q6c-ii
- 1 $(\text{C}_4\text{H}_9)^+$ cie-9701-2025-mj-p24-q6c-iii

Overlap with Paper A

Shared CIE IDs (same stem): 2025-mj-p12-q14, 2025-on-p11-q40, 2021-mj-p11-q7, 2025-mj-p11-q4.

New structured: B1 = 2025 MJ P22 Q2 (not P21 Q2); B2 = 2023 MJ P21 Q1(b)–(d) (not 2025 MJ P23 Q1; Table 1.1 deleted); B3 = 2025 MJ P24 Q6(c).

LO split: 1.1 = 6; 1.2 / Ar = 5; 1.3 = 6; 1.4 = 12; 22.2 MS = 4; total 33.